

Design of High Sensitivity Interdigital Capacitive Humidity Sensor Based on Uncertainty Analysis

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ABSTRACT

Humidity sensors have been widely used in various fields of life, among which sensitivity is one of the most important indicators for its evaluation. Quantitatively analyzing the effect of structural errors on sensitivity is a big challenge in sensor design. To address this problem, this paper proposes a structural uncertainty analysis method. By establishing the mathematical model and uncertainty propagation model, the influence of the interdigital capacitive humidity sensor's structural uncertainty on the sensitivity is quantitatively analyzed. The accuracy and superiority of the proposed method are also verified by finite element simulation.

KEYWORDS

humidity sensor; high sensitivity; uncertainty analysis; finite element simulation

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1 INTRODUCTION

Humidity sensors have attracted much attention due to their wide application in various fields, including weather detection[1, 20, 23, 31], smart city[5, 6, 28, 36], smart living[3, 13, 26, 27, 30], industrial internet of things[2, 4, 12, 29], etc. As an important indicator for evaluating sensors, sensitivity reflects the output change caused by unit input change. In extreme environments such as low temperature and low humidity, the higher the sensitivity of the sensor, the greater the output change, which is helpful for signal processing. To develop humidity sensors with better performance, researchers have devoted a lot of effort to studying various sensing techniques, such as capacitive[9, 10, 16, 17], resistive[21], resonant[14], optical fiber[33], quartz crystal microbalance (QCM)[34] and surface acoustic wave (SAW)[32].

Interdigital capacitive humidity sensors have been mainly studied due to their low cost, high sensitivity, fast response, and easy integration characteristics. By measuring the change in the capacitance value of the structure, the relative humidity of the air can be calculated and the environmental humidity field can be constructed[7, 8]. Its structural parameters play an important role in both theoretical analysis and practical application of sensor sensitivity.

Many researchers have noted the importance of structural parameters for the humidity sensor design. Y Wang et al. established a mathematical model of the humidity sensor, analyzing the influence of the interdigital plate spacing and the thickness of the humidity-sensitive layer on the sensor capacitance[25]. A. Tetelin et al. studied the influence of interdigital structure width and spacing on the dynamic response of the sensor[22]. H Liu et al. constructed a finite element model of the capacitive humidity sensor and the influence of some geometric parameters of humidity sensor on sensitivity is clarified[19].

However, the effect of the sensor structure on the performance analyzed in the above research work is still in the qualitative analysis stage. For low-cost, small-sized sensors, the quantitative impact

of errors in each structure during fabrication or measurement on sensitivity is unclear. Using the uncertainty analysis method can accurately guide sensor structure design, optimize sensor fabrication techniques, and minimize the effects of fabrication errors. At the same time, it can replace the traditional finite element analysis method to save a lot of time and resources, and provide fast and flexible guidance.

In this paper, we established a propagation model of the interdigital capacitive humidity sensor structure uncertainty, quantitatively analyzed the influence of structure error on sensitivity, and verified the accuracy of the method through finite element simulation analysis. On the one hand, based on the electrostatic field analysis method, the spatial potential distribution of the sensor is calculated, and the corresponding relationship between the sensor structure parameters and the sensitivity is obtained. On the other hand, the Monte Carlo sampling method is used to combine randomly selected structural parameters into numerous samples to propagate input parameter uncertainty. By calculating the discrete degree of the output sensitivity, the influence of the slight variation of each structural parameter on the system performance is quantitatively analyzed. The contributions of this paper are as follows:

- Establish a mathematical model of the interdigital capacitive humidity sensor, which provides theoretical support for the quantitative evaluation of factors affecting the sensitivity of the sensor.
- Propose an uncertainty analysis method, which quantitatively analyzes the influence of the structural errors generated in the process of machining or measurement on the sensitivity performance of the sensor.
- Built a finite element simulation model of an interdigital capacitive humidity sensor, which verifies that the accuracy of the proposed uncertainty analysis method reached more than 80%, and the speed is increased by more than 500 times compared with the simulation analysis.

The rest of this paper is organized as follows. Mathematical modeling is presented in Section 2. Section 3 describes the uncertainty analysis proposed in this paper. The last two parts, Section 4 and Section 5, are evaluation and conclusion respectively.

2 MATHEMATICAL MODELING

The interaction relationship between the structure and sensitivity of the sensor is obtained by modeling and analyzing the electric field of the sensor. In uncertainty analysis, the mathematical model can be used to convey the uncertainty of each structural parameter. Figure 1(a) shows the basic structure of an interdigital capacitive humidity sensor. A pair of interdigitated electrodes cross each other to form an electrode layer placed on the substrate, and the moisture-sensitive layer evenly covers the surface of the electrode[24]. When the relative humidity of the environment changes, the dielectric constant of the moisture-sensitive layer will change and affect the electric field of the sensor, as shown in Fig 1(b). The relative humidity of the environment can be calculated by detecting the change of capacitance value. Fig 1(c) shows several variable parameters of the interdigital capacitive humidity sensor. The length and width of the interdigital structure are denoted by L and b respectively, the distance between the two electrodes is denoted as a , the horizontal

distance between the electrodes is Δx , and the thickness of the moisture-sensitive layer is d .

In the process of modeling the electric field of the sensor, to simplify the problem, only a pair of interdigitated structures is taken as the research object, and the other structures are calculated in parallel. The electric field model between the two electrodes is established, as shown in Fig 1(d). Suppose they are on the same plane, and the potentials are U_1 and U_2 , respectively. α is the angular change caused by the thickness of the dielectric. U_1' and U_2' are the upper surface potentials of the moisture-sensitive layer dielectric. ϵ , ϵ_0 , ϵ_a are respectively the relative permittivity of moisture-sensitive layer, air, and substrate.

According to the symmetry of the charge distribution, the electric potential of each point on the central symmetry plane of a pair of electrodes is equal, and the electric field strength of each point is perpendicular to this plane. The potential of each point is only related to the angle θ , and the electric field strength is only related to the radius vector $\vec{r}$ from the origin to the point. Since there is no charge in the empty area between the electrodes, the space potential distribution should satisfy the Laplace equation

$$\nabla^2 U = \frac{\partial^2 U}{\partial r^2} + \frac{1}{r} \frac{\partial U}{\partial r} + \frac{1}{r^2} \frac{\partial^2 U}{\partial \theta^2} = 0 \quad (1)$$

There is an electric field above and below the electrodes. Through the boundary conditions on the upper surface of the electrode, the relationship between the upper surface potential and the angle can be calculated

$$U(\theta) = \begin{cases} U_1' + \frac{U_1' - U_1}{\alpha} \theta & \theta \in [-\alpha, 0) \\ U_1' + \frac{U_2' - U_1'}{\alpha} \theta & \theta \in [0, \pi] \\ U_2' - \frac{(U_2' - U_2')\pi}{\alpha} + \frac{U_2' - U_2'}{\alpha} \theta & \theta \in (\pi, \pi + \alpha) \end{cases} \quad (2)$$

According to the relationship between electric field and electric potential $\vec{E} = -\nabla U = -\left(\frac{\partial U}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial U}{\partial \theta} \vec{e}_\theta\right)$, the electric field mode above the electrode can be obtained

$$E = \begin{cases} \frac{U_1 - U_1'}{r \arctan \frac{2d}{a+b}} & \theta \in [-\alpha, 0) \\ \frac{U_1' - U_2'}{r \arctan \frac{2d}{a+b}} & \theta \in [0, \pi] \\ \frac{U_2' - U_2}{r \arctan \frac{2d}{a+b}} & \theta \in (\pi, \pi + \alpha) \end{cases} \quad (3)$$

A Gaussian surface is created above the electrode. According to Gauss's theorem, the average charge density between electrodes can be expressed as

$$\begin{aligned} \sigma_r &= \int_0^\pi E d\epsilon \\ &= \left(\frac{U_1 - U_1'}{r \arctan \frac{2d}{a+b}} + \frac{U_2' - U_2}{r \arctan \frac{2d}{a+b}} \right) \epsilon_{pi} + \frac{U_1' - U_2'}{r\pi} \epsilon_0 \end{aligned} \quad (4)$$

The charged quantity Q on the half-space of the electrode is

$$Q = \int_{a/2}^{(a+b)/2} \sigma_r dS = L \int_{a/2}^{(a+b)/2} \sigma_r dr \quad (5)$$

Combined with the definition of capacitance, the capacitance of the space above the electrode is expressed as

$$C_U = \left\{ 2 \left[\frac{\epsilon AD}{B} \right]^{-1} + \left[\frac{\epsilon_0 AD}{\pi} \right]^{-1} \right\}^{-1} \quad (6)$$

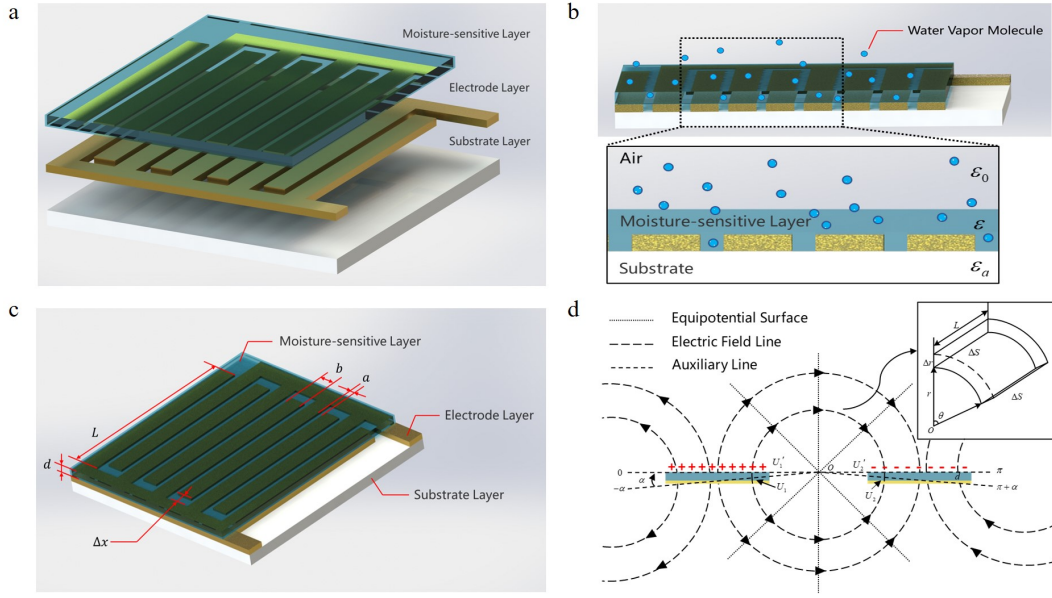


Figure 1: Interdigital Capacitive Humidity Sensor Structure

where $A = L - \Delta x$, $B = \arctan\left(\frac{2d}{a+b}\right)$ and $D = \ln\left(1 + \frac{b}{a}\right)$.

Similarly, the capacitance of the space below the electrode can be expressed as

$$C_L = \frac{\epsilon_a AD}{\pi} \quad (7)$$

Select polyimide (PI) as the moisture-sensitive material, which has the advantages of high sensitivity, linearity, and good hysteresis characteristics[11, 18]. After PI absorbs moisture, it forms a mixed medium with water vapor molecules. The dielectric constant of the mixed medium can be expressed as[15]

$$\epsilon = \left[\gamma_m \left(\epsilon_2^{\frac{1}{3}} - \epsilon_1^{\frac{1}{3}} \right) + \epsilon_1^{\frac{1}{3}} \right]^3 \quad (8)$$

where ϵ_1 and ϵ_2 represent the dielectric constants of PI and water, respectively, which are related to the temperature. γ_m is the volume percentage of vapor molecules in PI, which is related to relative humidity.

Therefore, the capacitance and sensitivity of the interdigitated capacitive humidity sensor based on PI material are

$$C = C_U + C_L = f_c(a, b, d, L, \Delta x, T, h) \\ = \frac{AD}{\frac{\pi}{\epsilon_0} + \frac{2B}{E^3}} + \frac{\epsilon_0 AD}{\pi} \quad (9)$$

$$S = \frac{dC}{dh} = f_s(a, b, d, L, \Delta x, T, h) \\ = \frac{6\gamma_m ABD [1 - \alpha(T - T_0)] \left(\epsilon_2^{1/3} - \epsilon_1^{1/3} \right) \psi(T)}{E^4 \left\{ \frac{\pi}{\epsilon_0} + \frac{2B}{E^3} \right\}^2} \quad (10)$$

where $E = \epsilon_1^{1/3} + \gamma_m [1 - \beta(T - T_0)] \left(\epsilon_1^{1/3} - \epsilon_2^{1/3} \right) h^{\psi(T)}$, β and $\psi(T)$ are the temperature coefficient and humidity index coefficient, respectively. h is relative humidity, $T_0 = 298K$.

3 UNCERTAINTY ANALYSIS

Based on the established electric field model of the interdigital capacitive humidity sensor, the uncertainty analysis of each structure is carried out. The influence of the processing or measurement error of each structure on the sensitivity is realized. Fig 2 shows the random modeling process of the input sample based on the uncertainty analysis. The structural parameters of the sensor, such as the length of the interdigital structure L , the width of the interdigital structure b , the distance a between the two electrodes, the thickness of the moisture-sensitive layer d and the horizontal distance between electrodes Δx , are selected as the uncertain input parameters. The samples of these structures are randomly extracted and combine into the model established in Section 2 to propagate the uncertainty of each parameter. By calculating the output distribution, the quantitative influence of each structure on the sensitivity is analyzed.

The input parameters $a, b, d, L, \Delta x$ are set to conform to the Gaussian distribution. The mean is the initial design value of each structural parameter, which is represented by μ , the standard deviation is represented by σ , and the COV (The coefficient of variance) is defined as σ/μ to measure the uncertainty of the input parameters. Each input parameter is randomly selected from a given Gaussian distribution using Monte Carlo sampling and combined into a single sample. Bring each randomly composed sample into the sensor electric field model to calculate sensitivity S_i

$$S_i = f_s(a_i, b_i, d_i, L_i, \Delta x_i), i = 1, 2, \dots, n \quad (11)$$

The distribution of the output sensitivity S_i for each sample combination is unknown. To quantify the uncertainty of the output parameters, the interquartile range (IQR) is used to reflect the degree of dispersion in the sensitivity. IQR is defined as the difference between the 25th percentile (P25) and 75th percentile (P75) [35].

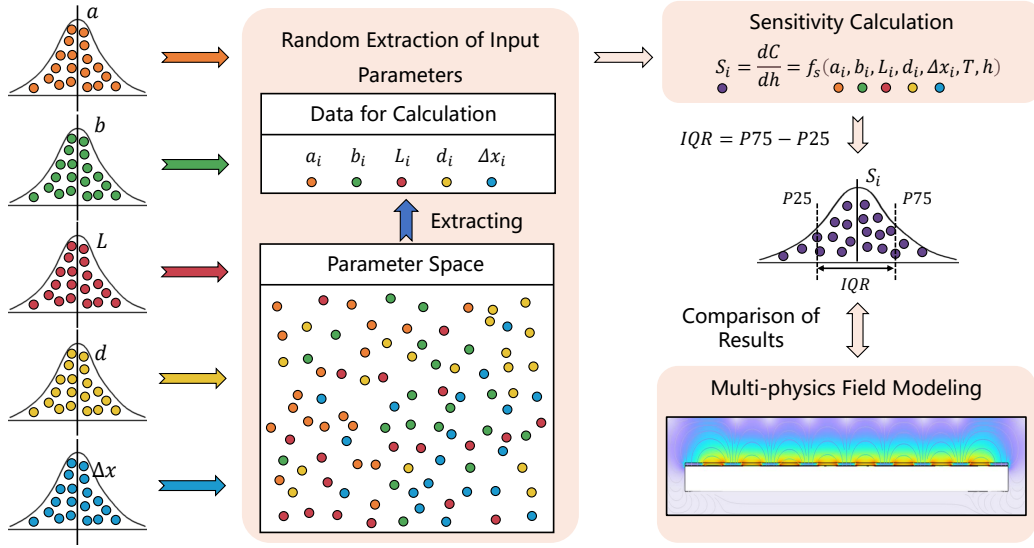


Figure 2: Theoretical Framework of Uncertainty Analysis

The larger the IQR value, the more scattered the data, and vice versa.

$$IQR = P75 - P25 \quad (12)$$

Considering the actual geometric size of the capacitive humidity sensor, the design values of the input parameter samples are set to $\mu_a = 5\mu\text{m}$, $\mu_b = 10\mu\text{m}$, $\mu_d = 3\mu\text{m}$, $\mu_L = 1250\mu\text{m}$, $\mu_{\Delta x} = 5\mu\text{m}$. Since there is a certain error in the processing or measurement of the sensor, the error variation coefficient COV is set according to the existing processing technology. To obtain an appropriate number of input samples, Figure 3 (a) and (b) show the convergence of the mean and standard deviation of the input parameters for different sample sizes. With the increase of the number of samples, the mean of each parameter converges rapidly, but the convergence of the standard deviation is still poor. When $N = 10000$, the fluctuation of standard deviation for input parameters of a , b , d , L , Δx converges to 0.76, 0.75, 0.56, 0.43, 0.93 %, respectively, which means the fluctuation is within a small range ($<1\%$). Set the number of input parameter samples to 10000 to propagate the uncertainty of structural parameters in the electric field model. The IQR of each structure is obtained from the histogram of the output sensitivity distribution, and the influence of each structural error can be quantitatively analyzed.

4 EVALUATION

The set initial conditions are brought into the model to calculate the influence of the uncertainty of the structurals on the sensitivity. COMSOL Multiphysics finite element analysis software is used to carry out comparative experiments in terms of accuracy, operation time and memory usage. All experiments were implemented on a computer equipped with an Intel Core i9-10920X and an NVIDIA GeForce 1660 SUPER.

The finite element analysis method can simulate complex engineering scenes as much as possible. Because of its standardization and normalization, it is the main tool in the research and design

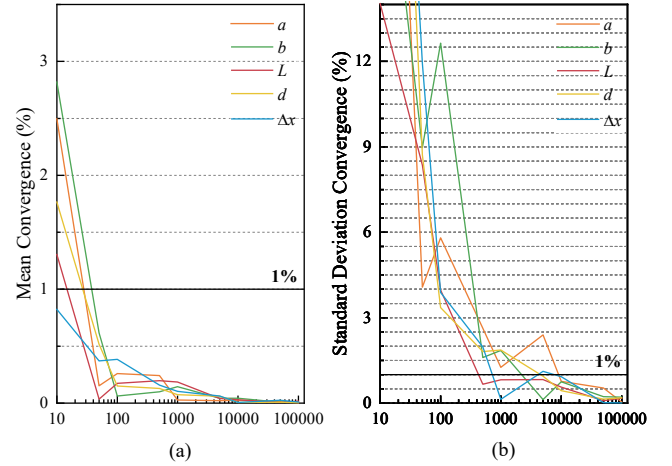


Figure 3: Convergence Analysis for Input Parameters

process. The finite element analysis software is used to build an interdigital capacitive humidity sensor model, and the influence of slight structural changes on the sensitivity is analyzed. Based on the simulation results, Fig 4 shows the average accuracy of the uncertainty analysis method relative to the finite element analysis results when the COV varies in the range of 0.04 ~ 0.4. The model accuracy of the distance between the electrodes a , the width of the interdigital structure b , and the horizontal distance between the electrodes Δx is higher than 90%. The accuracy of the length of the interdigital structure L and the thickness of the moisture-sensitive layer d is slightly lower, respectively 87.05% and 81.96%. Due to the complex boundary conditions of the model in the finite element analysis, especially in the vicinity of the moisture-sensitive layer, there will be certain errors in the theoretical modeling. The relative error of less than 20% does not affect the judgment of the impact

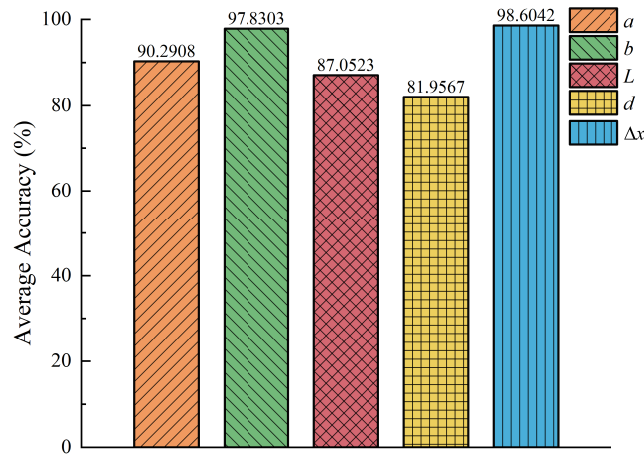


Figure 4: Computational Accuracy for Different Structures

of each structure on the sensitivity, which is within the acceptable range.

In terms of computing performance, Fig 5 (a) and (b) respectively compare the operation time and memory usage of uncertainty analysis and finite element simulation. After lots of experimental calculations, the average time-consuming of the finite element analysis method in the same computing environment is more than 500 times that of the uncertain analysis method. In contrast, the uncertainty analysis method is more flexible. It can immediately analyze the impact of structural uncertainty on sensitivity based on the given sensor parameters. The memory of the system occupied by the uncertainty analysis method is also less, which is 20% of that of the finite element analysis method.

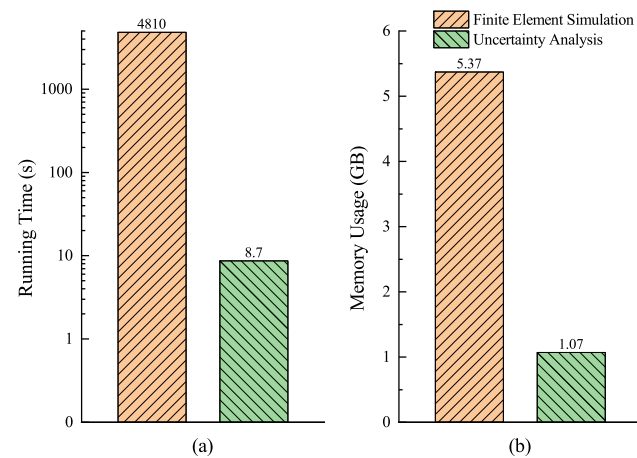


Figure 5: Comparative Experiment of Computational Performance

Applying the uncertainty analysis method, the functional relationship between the uncertainty degree IQR of the sensor sensitivity and the processing error COV of each structural parameter

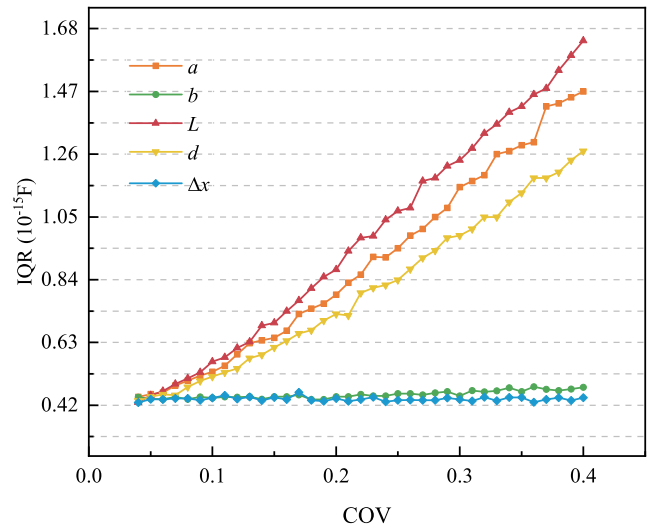


Figure 6: Effect of Sensor Structure Uncertainty on Sensitivity

a , b , d , L , Δx is shown in Fig 6. The variation range of COV is between 0.04 and 0.4. When the COV of a certain structural parameter changes, the COV of other parameters is a fixed value of 0.04. As can be seen from the figure, the IQR varies with COV and has a strong correspondence with the COV of each input parameter. Under the same COV conditions, the length L , the interdigital distance a , and the thickness d of the moisture-sensitive layer have a significant effect on the sensitivity of the sensor, while other structural parameters have no significant effect on the sensitivity. In the design and processing of the sensor of this structure, the processing accuracy of the above three structures should be mainly considered. The manufacturing accuracy of the interdigital width of the sensor can be appropriately sacrificed, but the glue-evening accuracy of the moisture-sensitive layer should be improved as much as possible.

5 CONCLUSION

In this paper, a sensor uncertainty analysis method is proposed, and the influence of each structural error of the interdigital capacitive sensor on the sensitivity is quantitatively analyzed. The method has been verified by finite element simulation that the accuracy of structural analysis is over 80%, and the operation time is shortened by more than 500 times. Through this method, we can not only quantify the influence of the uncertainty of each input parameter on the sensitivity but also greatly improve the experimental speed and flexibility compared with the traditional simulation analysis. At the same time, our method can be generalized to guide the design of sensors with different structures and sizes to improve sensor performance.

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